Multistatic Radar Simulator — System Architecture & Data Flow

E. Sai Shervan (CS24B073) · CS5013 · Packages under com.radar.simulator · same pipeline for the GUI and the CLI

1 · CONFIGURE

GUI form
`ui.ScenarioEditor`
TX, receivers (az/el), drone, timing, noise

Custom formulas (no code)
`ui.FormulaEditor → formula.FormulaSet`
parameters + formulas typed as text

Command line
`CLISimulatorApp`
--noise --seed --model --compare

Saved scenario file
`persistence.ScenarioPersistence`
scenario + formulas + chosen model

Build & check input
`ScenarioConfig.toScenario()`
`FormulaSet.compile()`
numbers, IDs, formula syntax

Scenario
`scenario.Scenario`
Transmitter + Receivers
+ TrajectoryModel

bad value

Invalid input
red message names the bad field
or formula (with character position)

2 · RUN THE EXPERIMENT

Geometry valid?
`localization.GeometryValidator`

no → **Refused**
explicit Status:
UNDER_CONSTRAINED
INVALID_INPUT
NUMERICALLY_UNSTABLE

yes (coplanar sensors → warning)

experiment.ExperimentRunner — repeat for every time step i (t = i · Δt)

① True drone position
`scenario.TrajectoryModel.getPosition(t)`
constant velocity: $p(t) = p_0 + v \cdot t$

② Simulate measurements
`measurement.MeasurementModel.generate()`
built-in: range = |p − TX| + |p − RX| (+ seeded noise), delay, power, az/el
or a user formula (FormulaMeasurementModel) · keyed by receiver ID

③ Estimate position (model chosen by the user)
`localization.LocalizationModel.estimate()`
Baseline LM least squares · Closed form · user formula (LM solver)
→ EstimationResult: position, Status, residual, condition no.

④ Compare with the truth
error = |true position − estimate|
(NaN if Status ≠ SUCCESS, excluded from metrics)

⑤ Record the step
`experiment.ExperimentRecord`
time, truth, measurement, estimate, error,
measurement + localization model name/version

next time step

Evaluate the run
`experiment.EvaluationModule → EvaluationResult`
mean / max / RMS error, localized vs failed steps

3 · SHOW & SAVE RESULTS

shown in red

GUI (ui.SimulatorController)
Visualization3D: sensors, true path, estimates, playback
ResultsPanel: per-step table · ComparisonPanel: models side by side
Status line: model versions, warnings, errors
`SceneTransform: East/North/Up → JavaFX axes`

CLI output
per-step table + summary, or --compare table

Save / export
`persistence.ScenarioPersistence`
save scenario + formulas (.properties) · export CSV
+ evaluation summary (.txt)

Replaceable formulations (interfaces)

`TrajectoryModel`	← ConstantVelocityTrajectory
`MeasurementModel`	← Baseline v1.1 · user formula
`LocalizationModel`	← Baseline v2.1 · ClosedForm v1.0 · formula

New formulation: type it in the GUI (no code), or add a class
+ one line in LocalizationModels. Name/version kept on every record.

Compare models (experiment.ModelComparison)
runs the loop once per model; measurements generated once
and replayed, so every model sees identical (noisy) data

Package colours: ui · formula · scenario · measurement · localization · experiment · persistence · errors / refused

dashed arrows = error path · ①–⑤ run once per time step